WORKED SOLUTION

STEP III 1991, Question 3

Pure • Piecewise Functions and Inverses

STEP III 1991, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q3

The key idea

A modulus is just a function that changes formula when its inside changes sign. The three moduli here change sign only at $x = -1$, 0 and 1, so on each of the four intervals they cut out of $x < 2$ the function is an ordinary quadratic. Once we have those four quadratics, every question (turning points, inflections, continuity, smoothness, inverse) is a question about simple parabolas and how they join.

Solution

Removing the moduli

Factorise the insides: $x^2 - x = x(x-1)$, $x^2 - 1 = (x-1)(x+1)$ and $x^2 + x = x(x+1)$. Their signs are:

interval	$x^2 - x$	$x^2 - 1$	$x^2 + x$	$f(x)$
$x \le -1$	$+$	$+$	$+$	$x^2 - 4x - 1$
$-1 \le x \le 0$	$+$	$-$	$-$	$3x^2 + 1$
$0 \le x \le 1$	$-$	$-$	$+$	$1 - 5x^2$
$1 \le x < 2$	$+$	$+$	$+$	$x^2 - 4x - 1$

For example, on $-1 \le x \le 0$,

$$f(x) = 2(x^2 - x) + (1 - x^2) + 2(x^2 + x) = 3x^2 + 1,$$

and on $0 \le x \le 1$, $f(x) = -2(x^2 - x) + (1 - x^2) - 2(x^2 + x) = 1 - 5x^2$. On the two outer intervals all three insides are positive, and $2(x^2 - x) + (x^2 - 1) - 2(x^2 + x) = x^2 - 4x - 1$. At the endpoints an inside is zero, so either formula is correct there.

Continuity and differentiability

f is built from $|\cdot|$ and polynomials, all continuous, so f is **continuous everywhere** on $x < 2$. As a check, the neighbouring formulas agree at the joins: $f(-1) = 4$, $f(0) = 1$,

$f(1) = -4$.

For differentiability, compare one-sided derivatives at the joins. The derivatives of the four pieces are $2x - 4$, $6x$, $-10x$ and $2x - 4$.

- At $x = -1$: from the left $2(-1) - 4 = -6$, from the right $6(-1) = -6$. These agree, so $f'(-1) = -6$.
- At $x = 0$: both sides give 0, so $f'(0) = 0$.
- At $x = 1$: from the left -10, from the right $2 - 4 = -2$. These differ, so the graph has a **corner** at $(1, -4)$.

So f is **not differentiable everywhere**: it fails exactly at $x = 1$.

Turning points and inflections

On every piece $f' < 0$, except at the single point $x = 0$:

- $x < -1$ and $1 < x < 2$: $f'(x) = 2x - 4 < 0$ since $x < 2$;
- $-1 < x < 0$: $f'(x) = 6x < 0$;
- $0 < x < 1$: $f'(x) = -10x < 0$.

Since f is continuous and decreases strictly on each piece, it is **strictly decreasing on the whole of** $x < 2$. In particular there are **no maximum or minimum points**, not even local ones.

? **Ask yourself** $f'(0) = 0$, so isn't $(0, 1)$ a turning point?

No. The left piece $3x^2 + 1$ has its minimum at 0 and the right piece $1 - 5x^2$ has its maximum there, but on the interval where each formula applies, both are decreasing. So f passes through $x = 0$ still going down, with a horizontal tangent.

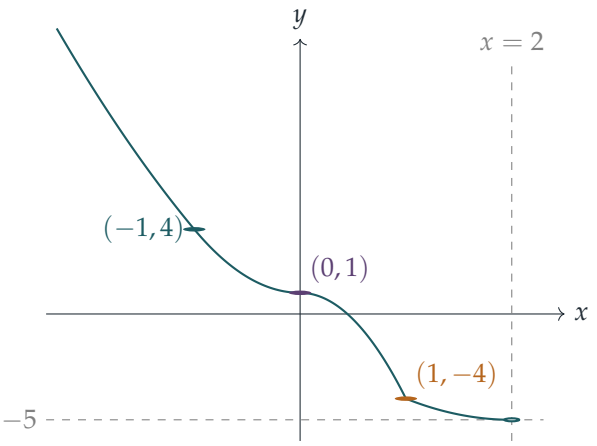
Now concavity. The second derivatives of the pieces are

$$f'' = 2 \ (x < -1), \qquad 6 \ (-1 < x < 0), \qquad -10 \ (0 < x < 1), \qquad 2 \ (1 < x < 2).$$

- At $x = 0$ the graph changes from convex to concave and has a tangent there, so $(0, 1)$ is a **stationary point of inflection**.
- At $x = -1$, f'' jumps from 2 to 6 but stays positive, so there is no inflection.
- At $x = 1$ the curve changes from concave back to convex, but it has a corner there and no tangent. By the usual definition (a tangent that crosses the curve) $(1, -4)$ is not a point of inflection. It's worth saying this explicitly in an answer.

Sketch

As $x \to -\infty$, $f(x) \to +\infty$. As $x \to 2^-$, $f(x) \to 4 - 8 - 1 = -5$, a value that is not attained.



The curve falls throughout. It flattens to a horizontal inflection at $(0,1)$ and has a corner at $(1,-4)$, where the slope jumps from -10 to -2.

The inverse

Because f is continuous and strictly decreasing on $x<2$, with $\mathrm{f}\to\infty$ on the left and $\mathrm{f}\to -5$ on the right, it is a bijection from $(-\infty,2)$ onto $(-5,\infty)$. So f^{-1} is defined on $x>-5$. We invert each piece on the range it covers, using $y=\mathrm{f}(x)$.

- $x\le -1$, range $y\ge 4$: $y=x^2-4x-1=(x-2)^2-5$, so $(x-2)^2=y+5$. Here $x-2<0$, so we take the negative root: $x=2-\sqrt{y+5}$.
- $-1\le x\le 0$, range $1\le y\le 4$: $x^2=(y-1)/3$, and $x\le 0$, so $x=-\sqrt{(y-1)/3}$.
- $0\le x\le 1$, range $-4\le y\le 1$: $x^2=(1-y)/5$, and $x\ge 0$, so $x=\sqrt{(1-y)/5}$.
- $1\le x<2$, range $-5<y\le -4$: again $(x-2)^2=y+5$ with $x-2<0$, so $x=2-\sqrt{y+5}$.

Renaming the variable as x:

$$\mathrm{f}^{-1}(x)=\begin{cases}2-\sqrt{x+5}, & -5<x\le -4,\\ \sqrt{\frac{1-x}{5}}, & -4\le x\le 1,\\ -\sqrt{\frac{x-1}{3}}, & 1\le x\le 4,\\ 2-\sqrt{x+5}, & x\ge 4.\end{cases}$$

The formulas agree at the shared endpoints. For example, at $x=4$ both give -1, and at $x=-4$ both give 1.

Checks

- Spot values: $\mathrm{f}(\frac{1}{2})=2\cdot\frac{1}{4}+\frac{3}{4}-2\cdot\frac{3}{4}=-\frac{1}{4}$, and $1-5\cdot\frac{1}{4}=-\frac{1}{4}$. Also $\mathrm{f}^{-1}(-\frac{1}{4})=\sqrt{1.25/5}=\frac{1}{2}$.

- Directly, $f(-3) = 2 \cdot 12 + 8 - 2 \cdot 6 = 20$. The formula gives $9 + 12 - 1 = 20$, and $f^{-1}(20) = 2 - \sqrt{25} = -3$.
- Numerically, $f^{-1}(f(x)) = x$ for 10^5 random $x < 2$, the function decreases at every point of a fine grid, and the one-sided slopes at $x = 1$ come out as -10 and -2.

Common slips

- Calling $(0, 1)$ a maximum or minimum because $f'(0) = 0$. Each parabola's own turning point is irrelevant if it sits at the edge of the interval where that formula holds.
- Picking the wrong square root in the inverse. The sign of x (or of $x - 2$) on each interval decides it.
- Assuming that a join between two formulas must be a non-differentiable point. At $x = -1$ and $x = 0$ the slopes match.

Beyond STEP

A piecewise-defined function can be continuous without being differentiable, and differentiable without having a continuous second derivative (here f'' jumps at $x = -1$ and $x = 0$). The inverse function theorem says that a strictly monotonic continuous function has a continuous inverse, and that the inverse is differentiable wherever $f' \neq 0$. Notice that f^{-1} has a vertical tangent at $x = 1$, exactly where $f'(0) = 0$. Abbott's *Understanding Analysis* develops these ideas carefully.

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